Vidyalankar Institute of Technology

Department of Computer Engineering

DEVOPS MINI PROJECT

Week 4 Report

Automated Grocery Inventory Dashboard

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1. Objective

The objective of Week 4 was to establish the Git and GitHub foundation for the Automated Grocery Inventory Dashboard.

The following activities were completed:

- GitHub repository creation.
- Spring Boot project initialization.
- Git configuration and initial commit.
- README and `.gitignore` setup.
- Branching strategy definition.
- GitHub issue templates.
- Remote repository connection and push.

This established the version-control foundation required for feature development and future DevOps activities.

2. GitHub Repository

The project repository was created on GitHub as the central location for source code and documentation.

Repository Name	automated-grocery-inventory-dashboard
Project Type	DevOps mini project for grocery inventory management
Purpose	Source code, documentation and version control
Development Model	Agile and DevOps-based development

Repository:

`github.com/Prachi5791/automated-grocery-inventory-dashboard`

3. Project Structure

The Spring Boot project was initialized using Java and Maven with a modular structure for future feature development.

The main structure contains:

The separation of controller, service, repository and model layers supports maintainable feature development.

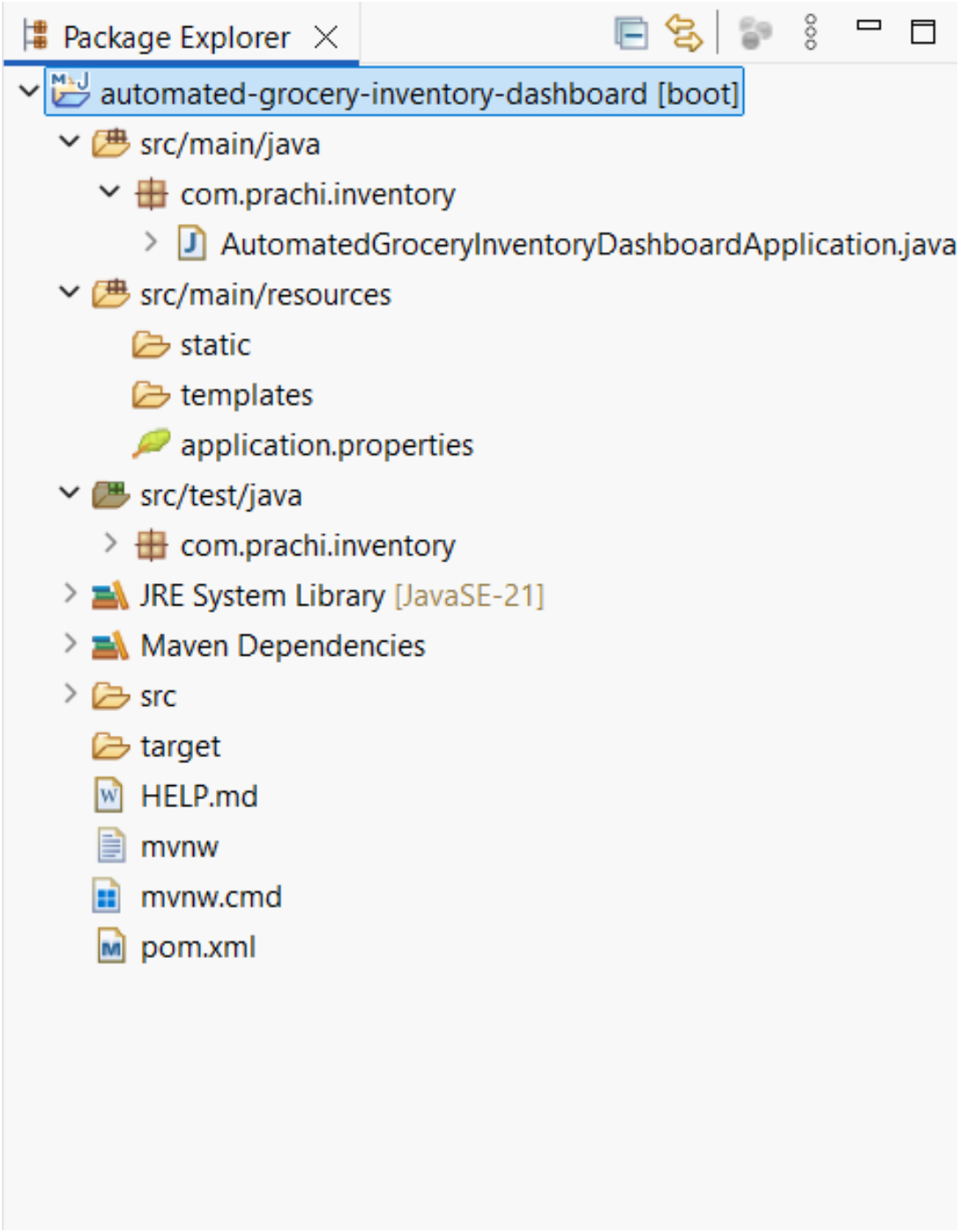


Figure 1: Project structure in the development environment

4. README Documentation

A README file was added to the project root to provide essential project information. It contains:

- Project overview and problem statement.
- MVP features.
- Technology stack.

- Project structure.
- Local setup information.
- Git workflow.
- Planned DevOps workflow.

5. Git Ignore Configuration

A `.gitignore` file was configured to prevent generated and environment-specific files from being committed.

Important entries include:

```
target/
.mvn/wrapper/maven-wrapper.jar
.idea/
.vscode/
*.class
*.log
.env
.DS_Store
```

The configuration keeps the repository clean by excluding build artifacts and IDE-specific files.

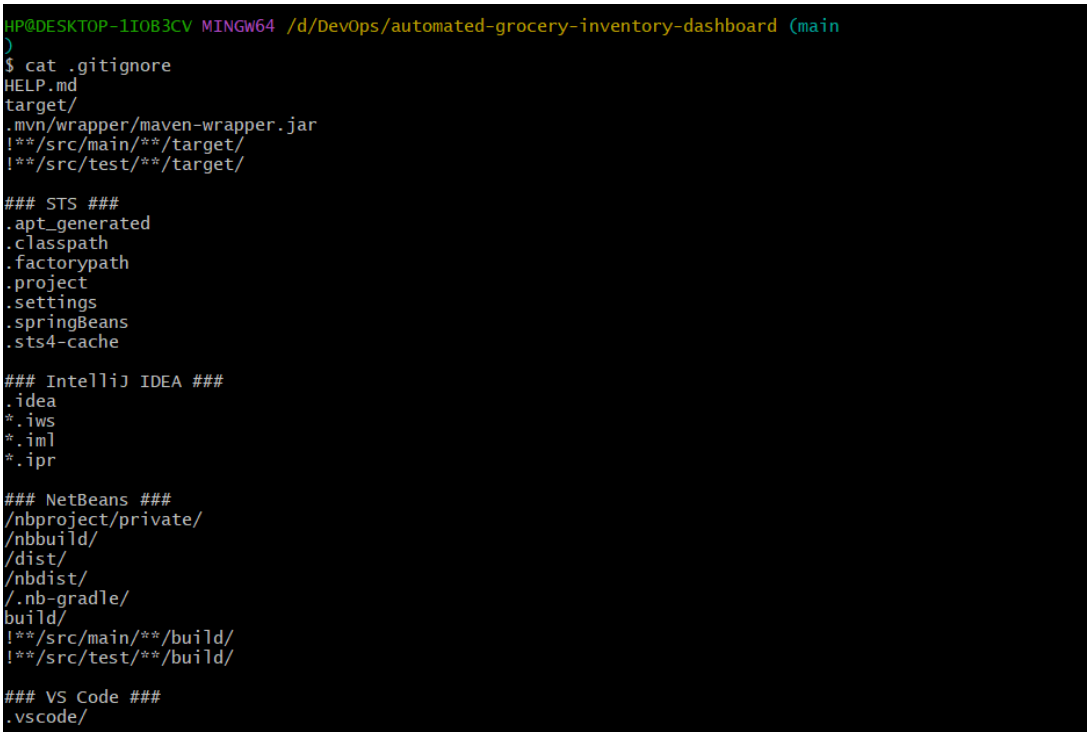


Figure 2: .gitignore configuration

6. Git Initialization

The local project was initialized as a Git repository and connected to the GitHub remote repository.

The main commands used were:

```
git init
git add .
git commit -m "Initial project setup with Spring Boot and Maven"
git branch -M main
git remote add origin <repository-url>
git push -u origin main
```

The initial commit established the first tracked version of the project.

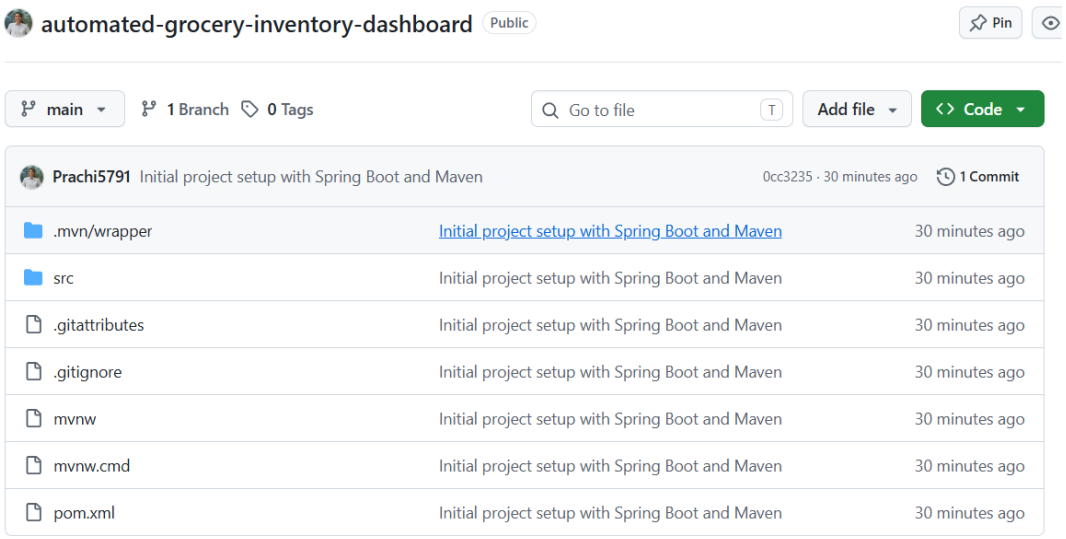


Figure 3: Git initialization and initial commit

7. Branching Strategy

A branching strategy was established to separate stable code from development and feature work.

Branch	Purpose
main	Stable and release-ready code.
develop	Integration branch for development work.
feature/*	Individual feature development.
bugfix/*	Bug-fixing work.

The development branch was created and the feature-branch approach was prepared for Week 5 implementation.

8. Branch Naming Policy

Consistent branch naming was defined for the project.

Branch Pattern	Purpose
main	Stable project code
develop	Development integration
feature/<name>	New feature development
bugfix/<name>	Bug correction

This naming convention was subsequently used for feature development.

9. GitHub Issue Templates

GitHub issue templates were configured to standardize project issue tracking.

Two templates were created:

- **Bug Report** – for errors and unexpected behavior.
- **Feature Request** – for new functionality or improvements.

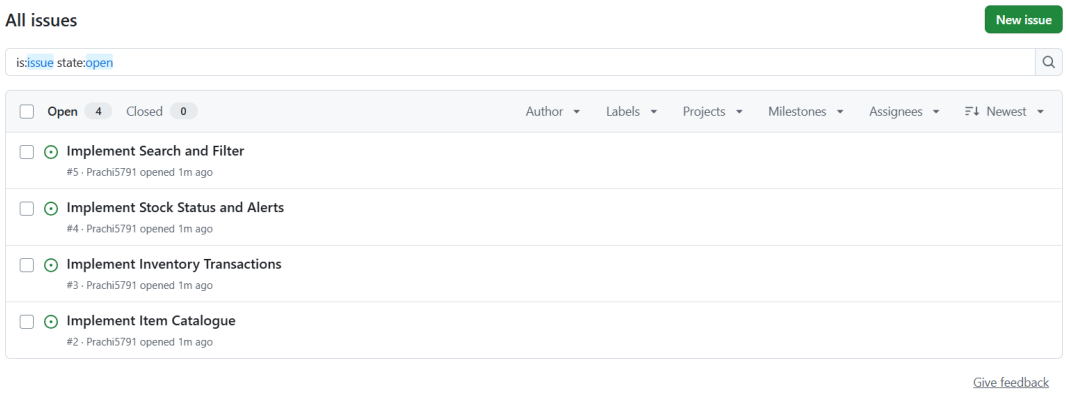


Figure 4: GitHub Issue Templates

10. Commit History

Meaningful commit messages were used to maintain a clear project history.

The initial project commit was:

```
Initial project setup with Spring Boot and Maven
```

Git history was verified using:

```
git status
git log --oneline
git branch
```

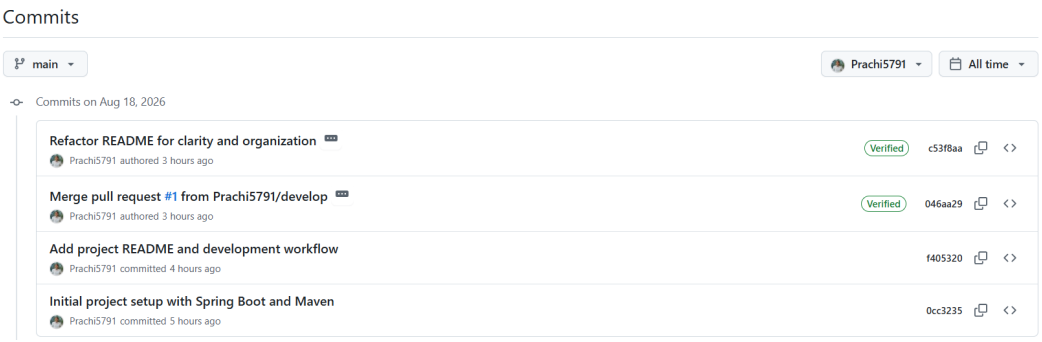


Figure 5: Git commit history

11. Week 4 Deliverables

The following deliverables were completed:

- GitHub repository created.
- Spring Boot project initialized.
- README documentation added.
- `.gitignore` configured.
- Local Git repository initialized.
- GitHub remote connected.
- Initial project pushed to GitHub.
- `develop` branch established.
- Branch naming strategy defined.
- GitHub issue templates created.
- Git history verified.

12. Learning Outcomes

Week 4 demonstrated:

- Git repository initialization.
- GitHub remote repository management.
- Git staging and commits.
- Branch creation and management.
- Branch naming conventions.
- Repository documentation.
- Git ignore configuration.
- GitHub issue management.

13. Conclusion

The Git and GitHub environment for the Automated Grocery Inventory Dashboard was successfully established.

The repository, project structure, documentation, Git configuration, branching strategy and issue templates provide the foundation for feature development in Week 5 and the subsequent DevOps workflow.

WEEK 4 – COMPLETED